

SEQ. BRK. ROUTINES (SBR0UT.55) 9/1/72 BPC/JBL

0/	OFFSET DCH	
7/	JMP DATACH	/1 DATA CHANNEL
13/	JMP READER	/2 READER
17/	JMP PRINTER	/3 LINE PRINTER
23/	JMP CONTRL	/4 CONTROLLERS
27/	JMP LD	/5 SWAPPING DRUM
33/	JMP TTSESV	/6 SCANNER
37/	JMP 1SEC	/7 1 SEC CLOCK
43/	JMP 1MIN	/10 1 MIN CLOCK
47/	JMP NETINT	/11 ARPA NETWORK INTERFACE
53/	JMP PUNCH	/12 PUNCH
57/	JMP PDP7	/13 PDP-7'S
63/	JMP SOR0BA	/14 TYPEWRITER
67/	JMP IOP	/15 IO PROCESSOR ISB'S
73/	JMP DISPAT	/16 RESTRICT MODE TRAP
77/	JMP SWAP	/17 32 MSEC. CLOCK

200/STAT.	TYEHNG	/STAT+177 MUST BE LESS THAN 8 BITS (377)
	REPEAT 77,	UNUSED

WHERE,	402	/XDDT IS IN CORE XCT
	REPEAT 77,300	/ALL OTHERS ARE UNUSED

+L

/TELETYPE TRANSLATION TABLES

600/ /TABLES START AT ODD MULTIPLE OF 200

/BIT 0 INDICATES 12 BIT CHAR. ON INPUT
 /BITS 1-6 ARE INTERNAL CODE OF INPUT CHAR.
 /BIT 7 INDICATES CHAR. IS CT ALARM CHAR.
 /BIT 8 INDICATES CHAR. IS NON-CT ALARM

/BITS 16-17 ARE TT OUTPUT CODE

W=402000 /WARNING ALARM CHARACTERS
 C=002000 /6 BIT CT ALARM CHARACTERS
 L=000000 /NON ALARM CHARACTERS
 U=000177 /OUTPUT CHARACTER FOR UNUSED INTERNAL CODES

DEFINE M TYPE,IN,OUT
 IN+T4000+TYPE+OUT
 TERMINATE M

TTTBL.	/INPUT CHAR	/OUTPUT CHAR
M L,00.040	/NULL	/SPACE
M W,01.041	/+A	/!
M W,02.042	/+B	/"
M W,03.043	/+C	/ #
M W,04.044	/EOT	/\$
M W,05.045	/WRU	/%
M W,06.046	/RU	/&
M W,07.047	/BELL	/i
M W,10.050	/+H	/(
M W,11.051	/TAB	/)
M W,12.052	/LF	/*
M W,13.053	/VT	/+
M W,14.054	/FORM	/.
M C,76.055	/CR	/-
M W,16.056	/+N	/.
M W,17.057	/+O	//
M W,20.060	/+P	/0
M W,21.061	/+Q	/1
M W,22.062	/TAPE	/2
M W,23.063	/XOFF	/3
M W,24.064	/+T	/4
M W,25.065	/+U	/5
M W,26.066	/+V	/6
M W,27.067	/+W	/7
M W,30.070	/+X	/8
M W,31.071	/+Y	/9
M W,32.072	/+Z	/:
363073	/"["=AM	/;
M W,34.074	/"BS"	/<<
M W,35.075	/"]"	/=
M W,36.076	/"+"	/>
M W,37.077	/"~"	/?
+L		

M C,00,100	/SPACE	/@
M C,01,101	/!	/A
M L,02,102	/"	/B
M C,03,103	/	/C
M C,04,104	/\$	/D
M C,05,105	/%	/E
M C,06,106	/&	/F
M C,07,107	/'	/G
M C,10,110	/(	/H
M C,11,111	/)	/I
M C,12,112	/*	/J
M C,13,113	/+	/K
M C,14,114	/,	/L
M C,15,115	/-	/M
M L,16,116	/.	/N
M C,17,117	//	/O
M L,20,120	/0	/P
M L,21,121	/1	/Q
M L,22,122	/2	/R
M L,23,123	/3	/S
M L,24,124	/4	/T
M L,25,125	/5	/U
M L,26,126	/6	/V
M L,27,127	/7	/W
M L,30,130	/8	/X
M L,31,131	/9	/Y
M C,32,132	/:	/Z
M C,33,133	/;	/
M C,34,175	/<	/AM
M C,35,135	/=	/I
M C,36,15	/>	/CRLF
M C,37,0	/?	

↑L

M C,40.000	/@	/NULL
M L,41.001	/A	/+A
M L,42.002	/B	/+B
M L,43.003	/C	/+C
M L,44.004	/D	/EOT
M L,45.005	/E	/WRU
M L,46.006	/F	/RU
M L,47.007	/G	/BELL
M L,50.010	/H	/+H
M L,51.011	/I	/TAB
M L,52.012	/J	/LF
M L,53.013	/K	/VT
M L,54.014	/L	/FORM
M L,55.015	/M	/CR
M L,56.016	/N	/+N
M L,57.017	/O	/+O
M L,60.020	/P	/+P
M L,61.021	/Q	/+Q
M L,62.022	/R	/TAPE
M L,63.023	/S	/XOFF
M L,64.024	/T	/+T
M L,65.025	/U	/+U
M L,66.026	/V	/+V
M L,67.027	/W	/+W
M L,70.030	/X	/+X
M L,71.031	/Y	/+Y
M L,72.032	/Z	/+Z
M C,73.033	/["	/["
M W,44.034	/BS	/BS"
M C,75.035	/]"	/]"
M W,46.036	/+"	/+"
M W,47.037	/+"	/+"

+L

/LOWER CASE ASCII INPUT DEFINED AS UPPER CASE

M L, 0. 0	/	/BREAK
M L, 41. 377	/A'	/RUBOUT
M L, 42. U	/B'	/
M L, 43. U	/C'	/
M L, 44. 134	/D'	/BS
M L, 45. U	/E'	/
M L, 46. 136	/F'	/↑
M L, 47. 137	/G'	/*

M L, 50. U	/H'	/
M L, 51. U	/I'	/
M L, 52. U	/J'	/
M L, 53. U	/K'	/
M L, 54. U	/L'	/
M L, 55. U	/M'	/
M L, 56. U	/N'	/
M L, 57. U	/O'	/

M L, 60. U	/P'	/
M L, 61. U	/Q'	/
M L, 62. U	/R'	/
M L, 63. U	/S'	/
M L, 64. U	/T'	/
M L, 65. U	/U'	/
M L, 66. U	/V'	/
M L, 67. U	/W'	/

M L, 70. U	/X'	/
M L, 71. U	/Y'	/
M L, 72. U	/Z'	/
M L, 0. U	/	/
M L, 0. 377	/	/RO
363000+U	/EOM	/
363000+U	/EOM	/
763000+U	/RO	/

TTTBL+200.
↑L

EXPUNGE C.L.M.W.U

1000.TTP. /TELETYPE PTRS. (200 WDS)

DEFINE TTPTRS NUM.ORG

REPEAT NUM.7

101 /LINE NOT OPEN. RNG SET

LAC 140000+[.-TTP-1]+T2+ORG /SERVICE PTR

LAC 140000+[.-TTP-2]+T2+ORG /USER PTR

0 /PGM WORD (0=DISCONNECTED)

7

TERMINATE TTPTRS

TTPTRS 10,100

/100-177 IS TT BUFFERS 0-7

TTPTRS 20,400-10+T10

/400-577 IS TT BUFFERS 10-27

TTPTRS 10,1200-30+T10

/1200-1277 IS TT BUFFERS 30-37

TTP/ 105

TTP+3/ STAT 0

1300/

WHAT.

201

/LD FIELD 0 SAVED FOR SWAPPING

REPEAT 37,400

/OTHER FIELDS EMPTY

201

/BD SLOT 0 SAVED FOR SWAPPING

REPEAT 61,400

/49. OTHER SLOTS

1422 CS1,

0

CS2,

200

/XDDT IS IN CORE XCT

CS3,

-3

/THERE AIN'T NO CORE DCH

SWFLDF.

0

SWFBDF.

40

RCORE.

0

ICORE.

0

DCORE.

100000

/XDDT IS IN PHYSICAL CORE 10

RSTAT.

0

ERSTAT.

DCH .

ISTAT.

0

WTOP.

WHERE 1

SWLGR.

0

/LAST GUY RUN. BUT NOT XDDT.

+L

/INITIALIZER

```

INIT.      LAC  .+2          /FLY THE SCANNER
           DAC  74
           CKS
           RIR  2S
           SPI
           RCC
           ISP  74
           JMP  INIT+2
           RTB
           CAC
           CBS
           LAW  DSWAP
           DAP  77
           ESM
           ISB  1700
           JMP  .
DSWAP.     LAW  SWAP
           DAP  77
           LIO  (340000)    /MEMORY PROTECT BITS
           ERM
           ASC  100
           ASC  200
           NOB
           ASC  400
           ASC  500
           ASC  600
           ASC  700
           ASC  1000
           ASC  1100
           ASC  1200
           ASC  1300
           ASC  1400
           ISB  600          /JUST TO MAKE SURE SCANNER FLYING
           ISB  1100         /  AND START UP NETWORK
           EEM
           JMP  I  .+1
DSWAPX.    DCH  SWORG       /DAC'ED INTO BY RESTART TAPE
^L

```

1500 /DATA CHAN SEQ BRK
 DATACH. LIF+USCF+UCLL
 DIO DATSAV
 1506 EEM
 JMP I DATADR

DATX. LIO DATSAV
 LFI
 LIO 6
 LAC 4
 JMP I 5

DATSAV. 0

/CONTROLLERS SEQ BRK
 CONTRL. LIO CONSAV
 LIF+ULFI
 DIO CONSAV
 EEM
 JMP I CONADR

CONX. LIO CONSAV
 LIF+ULFI
 DIO CONSAV
 LAC 20
 LIO 22
 JMP I 21

CONSAV. 0

/IO PROCESSOR LOW PRIORITY SEQ BRK
 IOP. LIF+USCF+UCLL
 DIO IOPSAV
 EEM
 JMP I IOPADR

IOPX. LIO IOPSAV
 LFI
 LIO 66
 LAC 64
 JMP I 65

IOPSAV. 0

/READER SEQ BRK
 DIMENSION RDRBUF(RDLEN)

READER. JSP RDR
 LIO 12
 LAC 10
 JMP I 11

RDVAR. DCH .
 RBI. COR RDRBUF
 RBO. COR RDRBUF
 RBE. COR RDRBUF+RDLEN
 RBK. 0
 RDWHO. -0
 +L


```

RDR,      RRB
RB1,      XX          / RPA OR DIO RB1
          DAP READEX
          LAC RBT
          SMA
          JMP RB4
          LAC I RBT
          RCL 9S
          LAC RBT
RB4,      RIL 9S
          DIO I RBT
          ADD (400000)
          SAD RBE
          SUB (RDLEN)
          DAC RBT
          ADD (1)      /NOT IDA: MIGHT BE IN RING MODE.
          SUB RBE
          RAL 1S
          SMA
          SUB (RDLEN RDLEN)
          RAR 1S
          ADD RBE
          SUB RBO
          RAL 1S
          SPA
          ADD (RDLEN RDLEN)
          SZA I
          JMP RB6
          SAS (RDLEN RDLEN-40)
          JMP RB5
7RDRI,    LAC RDWHO
          SPA          /OR SAD (-0)
          JMP RB5      /NO USER
          SAD BDSTAT
          LAW FSTAT
          DAP RDVAR
          LIO (3
          LAW I 7777   /BUFFER FILLING
          LSM
          AND I RDVAR
          SAS (RPAHNG) /HUNG ?
          JMP RB7      /NO
          DIO I RDVAR
          DZM HOTFLG
RB7,      ESM
RB5,      RCK
          DIO RBK
READEX,   JMP .
RB6,      LIO .+1      /BUFFER FULL ON NEXT CHARACTER
CRB1,     DIO RB1
          JMP RB5
+L

```

/PUNCH SEQ BRK

DIMENSION PUNBUF(PUNLEN)

PUNCH. / IDX PRUN

LAC PSP

SAD PUP

JMP PUNX

- DZM PRUN

LIO I PSP

PPA

IDX PSP

SAD PEND

SUB (PUNLEN)

DAC PSP

ADD (10)

SAD PUP

JMP PUNALM

SUB (PUNLEN)

SAS PUP

/10 CHARS FROM FULL?

JMP PUNX

PUNALM.

LAC PUNWHO

SPA

/OR SAD (-0)

JMP PUNX

/NO USER

SAD BDSTAT

LAW FSTAT

DAP PVAR

LAW I 7777

LSM

AND I PVAR

SAS (PPAHNG)

/HUNG ?

JMP PUNCH1

/NO

LAW 3

DAC I PVAR

DZM HOTFLG

PUNCH1.

ESM

PUNX.

LIO 52

LAC 50

JMP I 51

PUNWHO.

-0

PSP.

COR PUNBUF

PUP.

COR PUNBUF

PEND.

COR PUNBUF+PUNLEN

PVAR.

DCH .

PRUN.

.

+L

/SOROBAN SEQ BRK

```
SOROB.  EEM
        LSM
        JSP I SORADR
SORX.   ESM
        LIO 62
        LAC 60
        JMP I 61
```

/1 SEC CLOCK

```
1SEC.   LIO (1)           /CHANNEL 7 ROUTINE TO UNHANG
        LAW STAT          /USERS WHO ARE CLOCK HUNG
1SECL1. DAP 1PTR
1SECL2. LAW I 7777        /LOOP TO SEARCH FOR CLOCK-HUNG
        LSM
1PTR.   AND .
        SAD (CLKHNG)
        JMP 1SECT2        /FOUND ONE
1SECT1. ESM
        IDX 1PTR
        SAD (AND FSTAT+1)
        JMP HUNG
        ADD (WHERE-STAT-AND)
        SAS WTOP
        JMP 1SECL2
        LAW FSTAT
        JMP 1SECL1

1SECT2. IDX I 1PTR        /COUNT TOWARD 777
        AND (777)         /SEE SPP & DDTWNT
        SZA               /TIME TO UNHANG?
        JMP 1SECT1        /NO
        DIO I 1PTR
        DZM HOTFLG
        JMP 1SECT1
```

↑L

RRI=IOT 37
RRO=IOT 17

TXHUNG: 702445 677756
HUNGFC, 123456
HUNGHS, 123456
HUNGTR, 123456
HUNGRRO, 0
HUNGCTR, -5

HUNG, EEM
LAC I XNUMRRO /INDEXED ON EVERY "RRO+500"
SAD HUNGRRO
JMP HUNG2
DAC HUNGRRO
LAW I 5
DAC HUNGCTR
JMP 1SECC
HUNG2, LSM
RRI
LAI
AND (400027)
SAD (400021) /ABN,BSY,CONNECTED ?
ISP HUNGCTR
JMP HUNG3
LIO (700000)
RRO+500
NOP /DIAG SWITCH UP ?
LAC HUNGCTR
SZA
JMP HUNG3
RRI+100
DIO HUNGFC
RRI+1000
DIO HUNGHS
RRI+1100
DIO HUNGTR
LAC (TXHUNG)
JDA 14STXT /"HUNG" ON SOROBAN
HUNG3, ESM

↑L

1SECC, LSM
LAC 0CTR
DZM 0CTR
ESM
SZA
JSP I 1SECJA

1SECA, LSM
LAC 1CTR
DZM 1CTR
ESM
SZA
JSP I 1SECJB

1SECB, SZS I 40
7TT2, JSP .+1
LAC RB1
SAS RB2
JMP 7RDRX

/NORMAL .+1; ABNORMAL 7TT4. SET ON CH6

LSM
RCK
LAI
SUB RBK
ESM
SPA
ADD (60000.)
SUB (100.)
SPA
JMP 7RDRX
LIO .+1
DIO RB1
DSC 200
LAW 7RDRXX
DAP READEX
JMP 7RDRI

7RDRXX, ASC 200
7RDRX, LIO 36
LAC 34
JMP I 35

0CTR. 0
1CTR. 0
/1 MINUTE CLOCK
+L

```
1MIN.      EEM
            JMP I 1MIADR
1MIX.      LIO 42
            LAW 1000
            DAP TTF      /SEE SWNN DISPLAY HACK
            LAC 40
            JMP I 41
```

```
/LINE PRINTER INTERRUPTS (CHANNEL 3?)
PRINTER,   EEM
            JMP I PTRBRK
```

```
PTRRTN.    LIO 16
            LAC 14
            JMP I 15
```

```
/PDP-7 INTERRUPTS (CHANNEL 13)
PDP7.      EEM
            JMP I TELINT
PDP7X.     LIO 56
            LAC 54
            JMP I 55
```

```
/NETWORK INTERRUPTS (CHANNEL 11)
NETINT.    EEM
            JSP I NTINT
            ESM
            LAC 44
            LIO 46
            JMP I 45
```

```
START
↑L
```